

AI-Assisted Software Development

Mastering AI Code
Assistants

Mike Burton
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KnowledgeSolutions

Welcome to the Course

Course Objectives

At the end of this course, you will be able to:

- Effectively utilise AI code assistants like Cursor and GitHub Copilot in your daily development workflow
- Write optimal prompts to generate high-quality code and documentation
- Integrate AI-powered tools for testing, refactoring, and code analysis
- Apply best practices for reviewing and customising AI-generated code

Prerequisites

For this course, you need:

- Good proficiency in Python, Java or TypeScript
- Familiarity with modern software development practices and tools
- Experience with version control systems (e.g., Git) and IDEs



Why Study This Subject?

Mastering AI Code Assistants is vital:

- AI code assistants are revolutionising software development
- An essential tool in modern development workflows
- Mastering these tools can significantly boost your productivity and code quality
- Stay ahead in your career by developing expertise in this rapidly evolving technology

About Me

Mike Burton

- Software Development Trainer & Consultant
- Extensive development experience and a lover of AI-assisted development
- Passionate about teaching and technology

About you...

- ✓ Your name
- ✓ Your job
- ✓ Your experience
- ✓ Your expectation for the course



How We Will Work Together

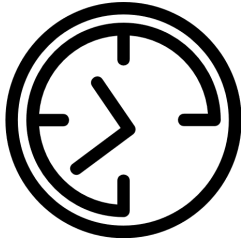
- Complete course materials provided (PDF, DRM-free) and lab exercises
- Interactive sessions with concepts, demonstrations, and hands-on practice
- Individual and group exercises
- Real-time feedback and discussion
 - *Ask questions!*
 - *Interrupt me at any time*

My Aims

I will aim to:

- Create an interactive and engaging learning environment
- Encourage active participation and questions
- Provide practical, real-world examples
- Ensure everyone gains hands-on experience

Logistics



Duration: 3 days

Schedule: 09:30 - 17:00 each day



Breaks: Morning, Lunch, Afternoon



Phones on silent!

Agenda

Overview of topics

- “AI” and how it can help software development
 - Cursor AI tool – for Coding
 - Other Tasks
 - Other Tools
- 
- ```
graph TD; A[Cursor AI tool – for Coding] --> B[Other Tools]; A --> C[Other Tasks];
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A man in a blue checkered shirt is standing and gesturing while speaking to a group of people. The group, consisting of a woman sitting on a wooden table and several people standing, is listening attentively. The setting is a bright room with large windows in the background. The text "Any Questions?" is overlaid in the upper center of the image.

# Any Questions?

The logo for Knowledge Solutions, featuring the letters "kpi" in a bold, lowercase, sans-serif font inside a white square.

kpi

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